MPG No. 8.7.126 $m'mm$ (mm'm setting) [Type III, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ M_{xz} & 0 & 0 \end{bmatrix}$$

$M_{xz} = \mathbb{M}^{(1)}_{xz}$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ -M_{xz} & 0 & 0 \end{bmatrix}$$

$M_{xz} = -\mathbb{T}^{(1)}_{y}$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ M_{xxzx} & M_{yyzx} & M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzxy} & 0 & 0 \end{bmatrix}$$

$M_{xxzx} = -\mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz}$
$M_{yyzx} = 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz}$
$M_{zzzx} = \mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz}$
$M_{yzxy} = 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(2)}_{xz}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ -M_{xxzx} & -M_{yyzx} & -M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$M_{xxzx} = \mathbb{M}^{(3)}_{xz} - 2\mathbb{T}^{(1)}_{y} + 2\mathbb{T}^{\alpha(1)}_{y} + 2\mathbb{T}^{\beta(1)}_{y}$
$M_{yyzx} = \mathbb{M}^{(3)}_{xz} - 4\mathbb{T}^{\beta(1)}_{y}$
$M_{zzzx} = \mathbb{M}^{(3)}_{xz} + 2\mathbb{T}^{(1)}_{y} - 2\mathbb{T}^{\alpha(1)}_{y} + 2\mathbb{T}^{\beta(1)}_{y}$
$M_{yzxy} = \mathbb{T}^{(1)}_{y} + 4\mathbb{T}^{\alpha(1)}_{y}$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ M_{yzxy} & 0 & 0 \end{bmatrix}$$

$M_{yzxy} = 2\mathbb{M}^{(4)}_{xz}$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = \mathbb{T}_y^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & 0 \\ 0 & M_{yyzx} & 0 \\ 0 & M_{zzzx} & 0 \\ 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ M_{xyyz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxzx} &= 2\mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(3)} - \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)} \\
M_{yyzx} &= \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{(4)} + 2\mathbb{T}_y^{\alpha(2)} \\
M_{zzzx} &= -2\mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(3)} - \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)} \\
M_{yzxy} &= \mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(4)} - \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)} \\
M_{xyyz} &= -\mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(4)} - \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzxy} &= -\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{zxxx} &= 2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{zxyy} &= \mathbb{T}_y^{(5)} + 2\mathbb{T}_y^{(6)} + 2\mathbb{T}_y^{\alpha(3)} \\
M_{zxzz} &= -2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{xyyz} &= \mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 & 0 & M_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 & 0 & M_{yyxz} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 & 0 & M_{zzxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzyx} \\ M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 & M_{xyzy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxzx} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} - 2\mathbb{T}_y^{(1)} + \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} - \mathbb{T}_y^{\beta(2)} \\
M_{xxxz} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} - 2\mathbb{T}_y^{(1)} - \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} + \mathbb{T}_y^{\beta(2)} \\
M_{yyzx} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} + \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{(4)} + 2\mathbb{T}_y^{\alpha(2)} - 4\mathbb{T}_y^{\beta(1)} \\
M_{yyxz} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} - \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{(4)} - 2\mathbb{T}_y^{\alpha(2)} - 4\mathbb{T}_y^{\beta(1)} \\
M_{zzzx} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} + 2\mathbb{T}_y^{(1)} + \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} + \mathbb{T}_y^{\beta(2)} \\
M_{zzxz} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} + 2\mathbb{T}_y^{(1)} - \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} - \mathbb{T}_y^{\beta(2)} \\
M_{yzxy} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(1)} + \mathbb{T}_y^{(4)} + 4\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)} \\
M_{yzyx} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(1)} - \mathbb{T}_y^{(4)} + 4\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)} \\
M_{zxxx} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} + 2\mathbb{T}_y^{(1)} - 2\mathbb{T}_y^{\alpha(1)} - 2\mathbb{T}_y^{\beta(1)} \\
M_{zxyy} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} - \mathbb{M}_{xz}^{(3)} + 4\mathbb{T}_y^{\beta(1)} \\
M_{zxzz} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} - 2\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)} - 2\mathbb{T}_y^{\beta(1)} \\
M_{xyyz} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)} - \mathbb{T}_y^{(1)} + \mathbb{T}_y^{(4)} - 4\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)} \\
M_{xyzy} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(5)} - \mathbb{T}_y^{(1)} - \mathbb{T}_y^{(4)} - 4\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzyx} \\ M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 & M_{xyzy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzxy} &= 2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(2)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} - \mathbb{T}_{y}^{\beta(3)} \\
M_{yzyx} &= -2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} - \mathbb{T}_{y}^{(2)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} - \mathbb{T}_{y}^{\beta(3)} \\
M_{zxxx} &= 2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(5)} - \mathbb{T}_{y}^{\alpha(3)} - \mathbb{T}_{y}^{\beta(3)} \\
M_{zxyy} &= \mathbb{T}_{y}^{(5)} + 2\mathbb{T}_{y}^{(6)} + 2\mathbb{T}_{y}^{\alpha(3)} \\
M_{zxzz} &= -2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(5)} - \mathbb{T}_{y}^{\alpha(3)} + \mathbb{T}_{y}^{\beta(3)} \\
M_{xyyz} &= 2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} - \mathbb{T}_{y}^{(2)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} + \mathbb{T}_{y}^{\beta(3)} \\
M_{xyzy} &= -2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(2)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} + \mathbb{T}_{y}^{\beta(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxz} \\ 0 & 0 & 0 \\ M_{zzzx} & 0 & 0 \\ M_{yyzx} & 0 & 0 \\ 0 & 0 & M_{zzxz} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{zxxx} & 0 & 0 \\ 0 & 0 & M_{xyyz} \\ 0 & M_{xyyz} + \frac{M_{zzxz}}{2} - \frac{M_{zzzx}}{2} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxz} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} \\
M_{zzzx} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} \\
M_{yyzx} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} \\
M_{zzxz} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} \\
M_{zxxx} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} \\
M_{xyyz} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)}
\end{aligned}$$